

Vedant Desai

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EDUCATION

University of Michigan

B.S. in Computer Science and Economics

Expected May 2028

Ann Arbor, MI

- **GPA:** 3.66 / 4.0
- **Coursework:** Data Structures & Algorithms, Intro to Statistics and Data Analysis, Discrete Mathematics, Calculus III

EXPERIENCE

Autonomous B2B Deal-Sourcing Agent

github.com/Verdent06/Lead-Gen-Agentic-Pipeline

Dockerized LangGraph pipeline for private equity acquisition sourcing — deployed across 4 live theses for search funds and mid-market PE clients

- Engineered a LangSmith eval pipeline with 20 adversarial business archetypes — manufacturers, SaaS tools, PE holding companies, and geographic mismatches the LLM otherwise passed — then layered deterministic Pydantic consensus gates to lift extraction faithfulness from 50% to 90%.
- Packaged a Dockerized LangGraph execution pipeline as a recurring B2B lead-gen product, cold-approached and closed a paid pilot with a consulting firm, and deployed Redis/Celery workers to autonomously aggregate and deliver qualified leads on a scheduled basis.
- Architected a custom asyncpg Data Access Layer storing OpenAI embeddings alongside source records; engineered injection-safe SQL timestamp validation that detects stale entries and triggers automatic re-scrapes, keeping the lead database fresh without manual intervention.

CaseStudyPrep.AI

Dec 2025 – May 2026

Software Engineer Co-op (Voice AI)

Remote

- Eliminated the silent-audio problem in a voice-AI product — most frames sent to the Whisper backend were dead air — by running Silero VAD client-side via ONNX Runtime, filtering silence before upload and cutting cloud inference costs by 40%.
- Eliminated a 27% audio upload failure rate by building fault-tolerant RxJS logic that detects expired S3 presigned URLs and regenerates them mid-flight, and handles MIME-type negotiation for raw WAV files that Angular's HTTP client rejected silently.

Michigan Data Consulting (MDC)

Jan 2026 – May 2026

Data Engineer — Michigan Campaign Finance Network

Ann Arbor, MI

- Replaced manual Michigan campaign-finance research — portal searches capped by the Bureau of Elections, irregular Excel exports, hand normalization at 2 hours per committee — with a Requests + Pandas ETL that ingests filings directly, eliminating 800 hours of manual pulls across 400 tracked PACs.
- Delivered a production Flask REST API on AWS EC2 as the sole engineer on a 5-month MCFN contract, wiring ingested data and PAC rankings into the nonprofit's public-facing research workflow.

PROJECTS

Granular Synthesizer Plugin | C++, JUCE, VST3, AU, CMake

github.com/Verdent06/granular-synth

Real-time audio DSP plugin built from scratch in C++/JUCE — sine oscillator through full granular engine with lock-free threading and release-ready VST3/AU binaries

- Configured CMake to compile VST3 and AU plugin bundles from a single JUCE codebase — targeting a macOS universal binary (arm64 + x86_64) so the plugin runs natively on Apple Silicon and Intel — and audited every processBlock() code path against a real-time safety checklist confirming zero heap allocations and zero lock acquisitions before cutting release builds.

TECHNICAL SKILLS

AI / ML LangGraph, LangSmith, Silero VAD, ONNX Runtime, Pandas
Languages Python, C++
Frameworks Flask, Angular, RxJS, JUCE
Cloud & Infrastructure AWS (EC2, S3), Docker, Redis, Celery, Git